

BUSINESS PERFORMANCE & OPERATIONS ANALYSIS

DataCo Global – Supply Chain & Retail

MIS 443- Data Management

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Group: DataMinds

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I. INTRODUCTION

BUSINESS BACKGROUND



DataCo Global – Supply Chain & Retail

- Multinational retail and supply chain company
- Operates across multiple markets and geographical regions
- Manages large volumes of:
 - Customer orders
 - Product sales
 - Inventory
 - Shipping & logistics data
- Data analytics supports operational efficiency and strategic decision-making

I. INTRODUCTION

BUSINESS PROBLEM

DataCo generates a large amount of transactional data, but raw data makes it difficult to:

- Identify profitable products
- Evaluate sales performance
- Analyze customer & regional demand
- Measure shipping efficiency
- Monitor overall profitability

Solution:

Relational Database + SQL + Python → Actionable Business Intelligence



I. INTRODUCTION

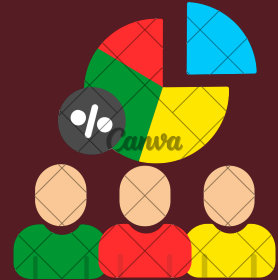
RESEARCH QUESTIONS & OBJECTIVES



1. Overall Business Performance

Question: How is the overall business performing in terms of sales, profit, and average order value?

Objective: Evaluate the company's overall financial performance.



2. Product Performance

Question: Which products and product categories contribute the most to sales?

Objective: Identify high-performing and low-performing products and categories.



3. Customer & Regional Analysis

Question: What are the purchasing patterns of different customer segments and geographical regions?

Objective: Understand customer purchasing behavior and identify the regions generating the highest sales.



4. Logistics & Delivery Performance

Question: How efficient is the company's logistics and delivery performance?

Objective: Evaluate shipping performance, delivery status, and delivery punctuality.



5. Sales & Profit Trends

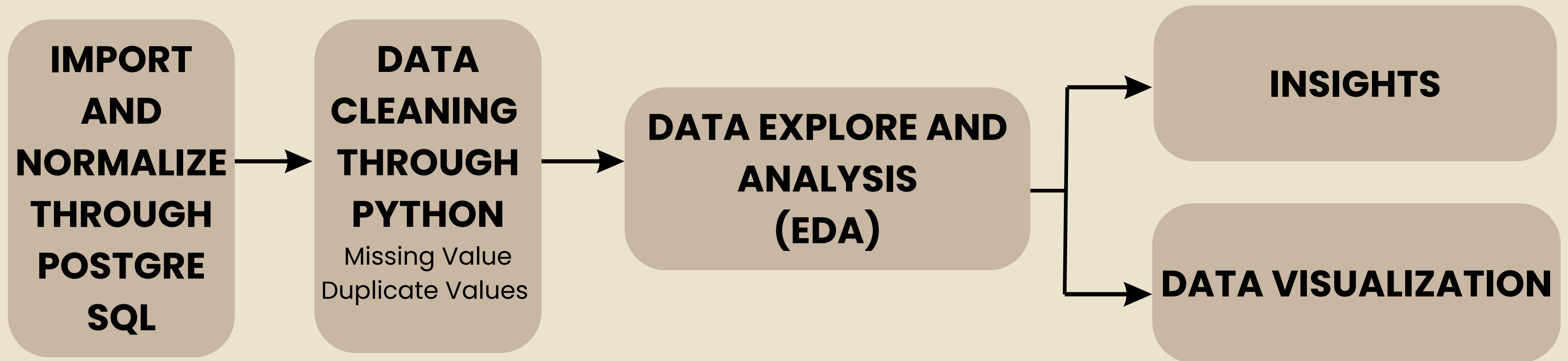
Question: How do sales and profit change over time?

Objective: Analyze monthly sales and profitability trends to support forecasting and business planning.

=>Provide data-driven recommendations to improve business and supply chain performance.

I. INTRODUCTION

DATA SOURCE FLOWCHART



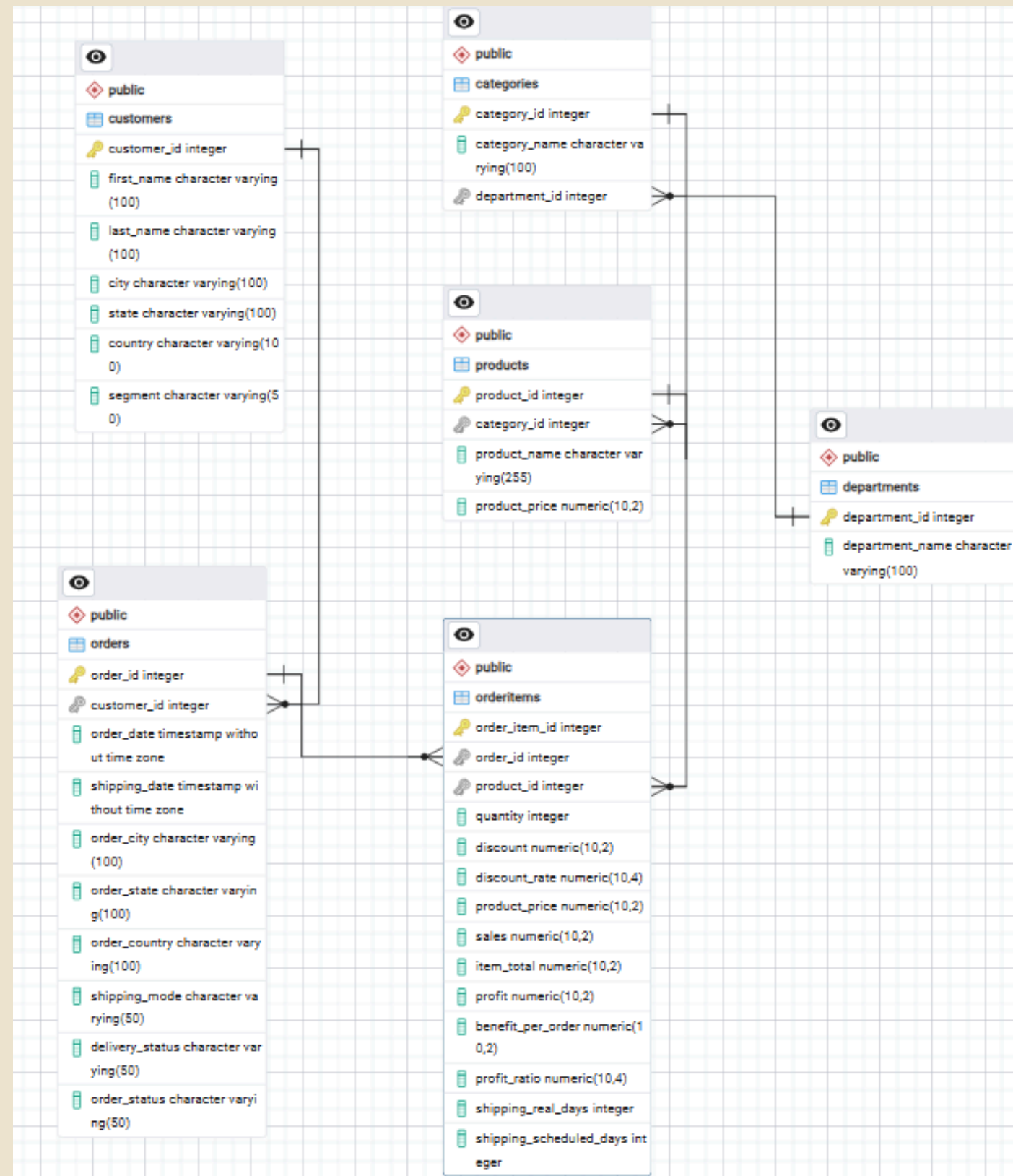
Dataset Overview

Attribute	Description
Dataset Name	DataCo Supply Chain Dataset
Source	Kaggle
Data Type	Structured transactional data
Number of Records	Approximately 180,000
Number of Attributes	50+
Business Domain	Supply Chain & Retail
Main Entities	Customers, Orders, Products, Categories, Shipping, Markets
Primary Purpose	Database implementation and business analytics



II. PROJECT OVERVIEW

Entity Relationship Diagram (ERD)



The relationships between these entities can be summarized as follows:

- One customer can place multiple orders.
- One order can contain multiple products.
- Each product belongs to one category.
- Each order uses one shipping method.
- Orders are associated with a specific market and geographical location.

II. PROJECT OVERVIEW

Tools and Technologies

Tool / Technology	Purpose
PostgreSQL	Relational database management system used to implement the database schema and store business data
pgAdmin 4	Graphical interface for database administration and SQL execution
SQL	Data definition, manipulation, and analytical querying
Python	Data extraction, preprocessing, analysis, and visualization
Pandas	Data manipulation and database connectivity
Matplotlib	Data visualization
Microsoft Excel	Initial inspection and verification of dataset quality
GitHub	Version control and project repository management

3. DATA ANALYSIS AND INSIGHTS



Research Question 1

HOW IS THE OVERALL BUSINESS PERFORMING IN TERMS OF SALES, PROFIT, AND AVERAGE ORDER VALUE?

Total Sales	Total Profit	Profit Margin	Average Order Value
\$36.78M	\$3.97M	10.80%	\$183.11

```
SELECT ROUND(SUM(sales), 2) AS total_sales
FROM OrderItems;

SELECT ROUND(SUM(profit), 2) AS total_profit
FROM OrderItems;

SELECT
    ROUND(AVG(item_total), 2) AS average_order_value
FROM OrderItems;
```

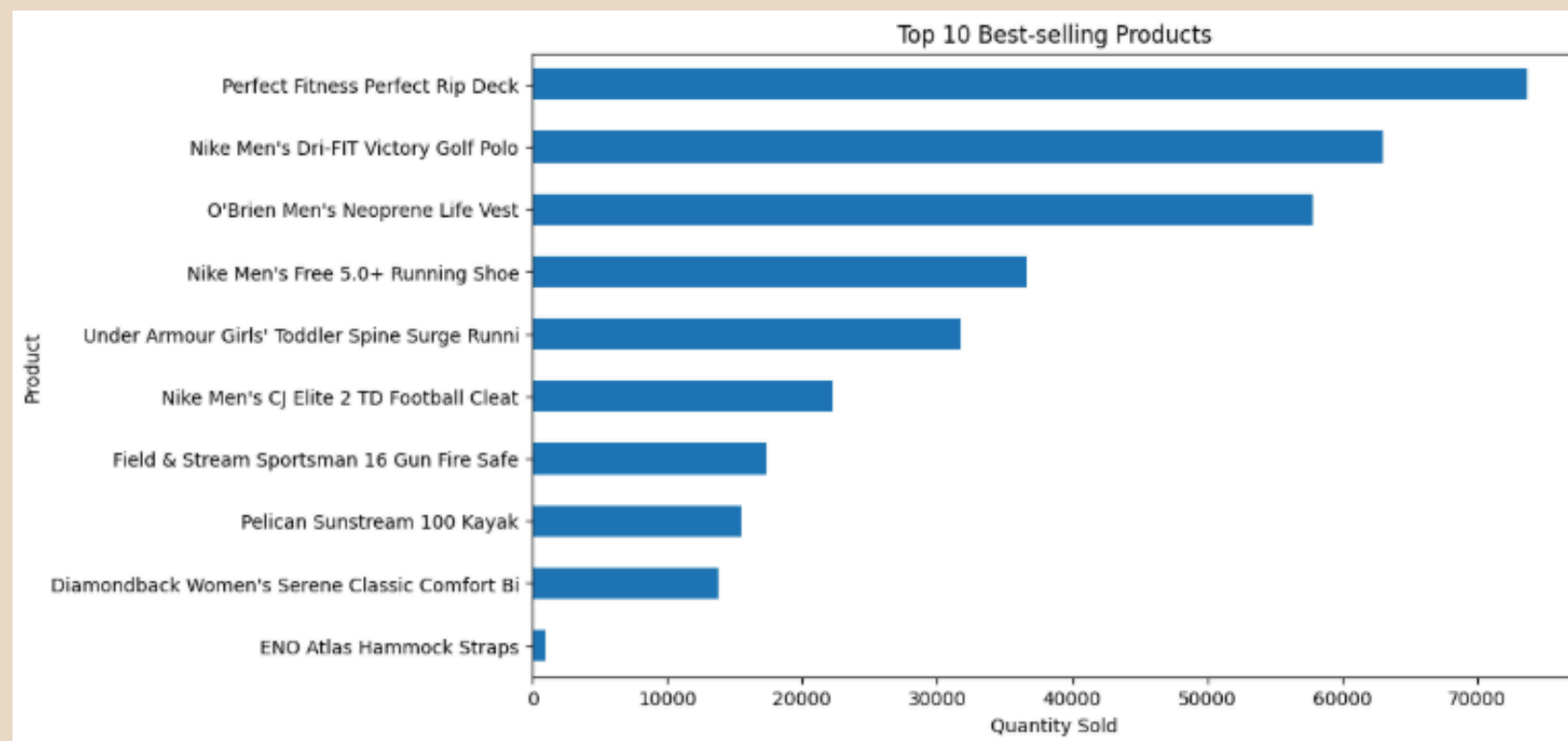
DataCo demonstrates healthy overall business performance, with strong sales and positive profitability.

- \$36.78M in sales indicates a strong revenue base.
- \$3.97M profit shows that the business generates positive returns.
- 10.8% estimated margin indicates room to improve profitability.
- \$183.11 AOV suggests customers typically purchase multiple or relatively high-value items per transaction.

PRODUCT PERFORMANCE

Research Question 2

WHICH PRODUCTS AND PRODUCT CATEGORIES
CONTRIBUTE THE MOST TO SALES?

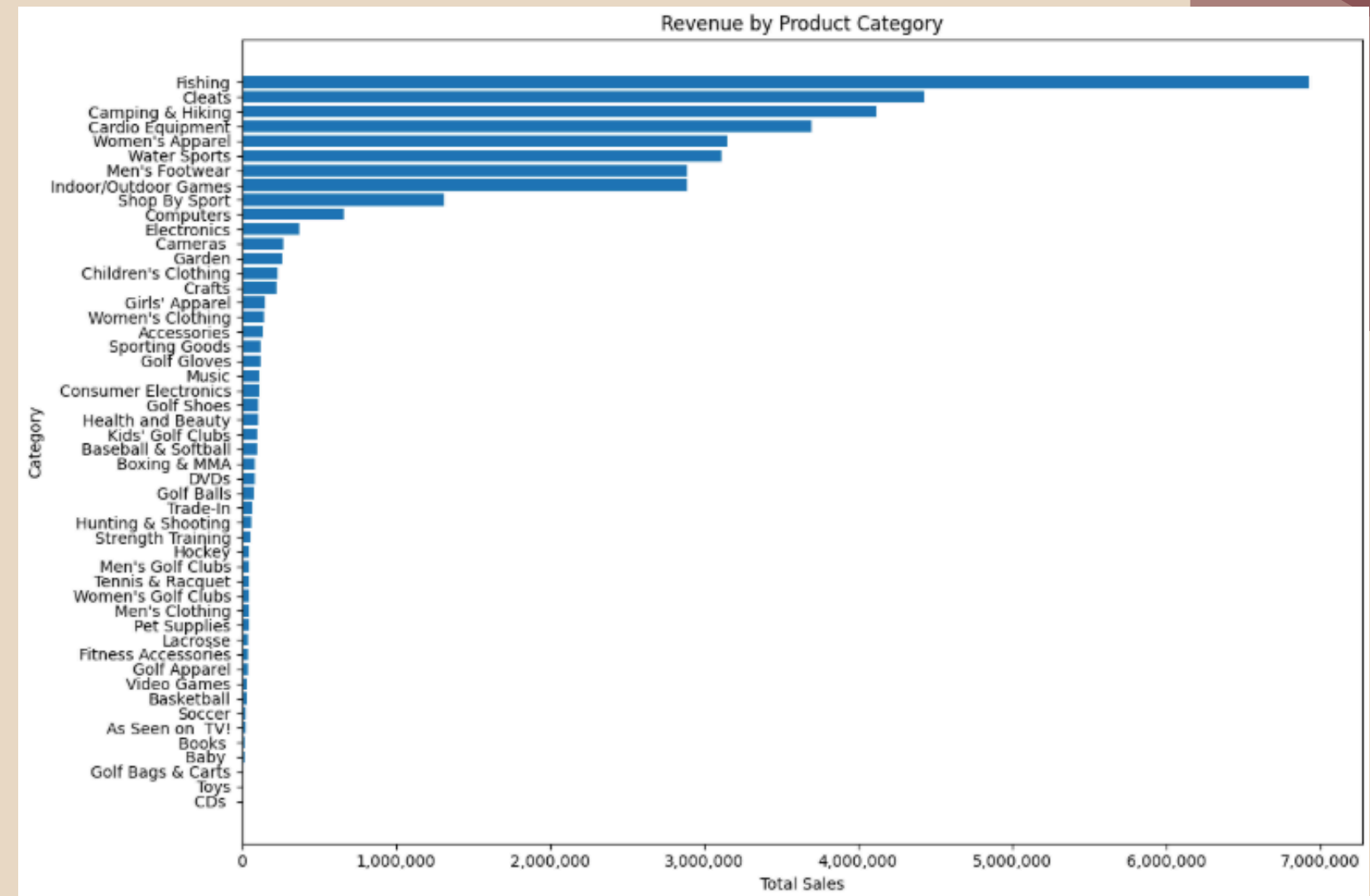


Perfect Fitness Perfect Rip Deck



Nike Men's Dri-FIT Victory Golf Polo

=> A small number of products dominate sales.



Fishing



Cleats



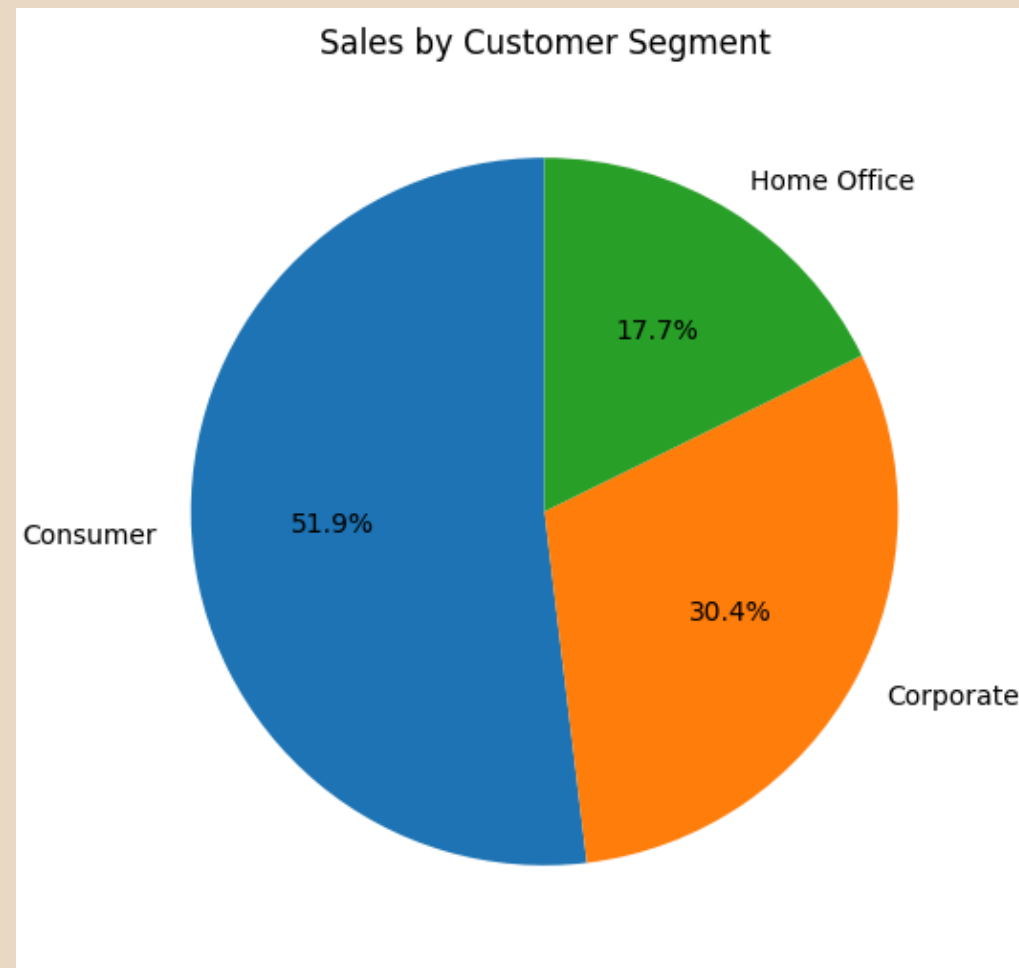
Camping & Hiking

=> Outdoor & sporting categories are key revenue drivers.

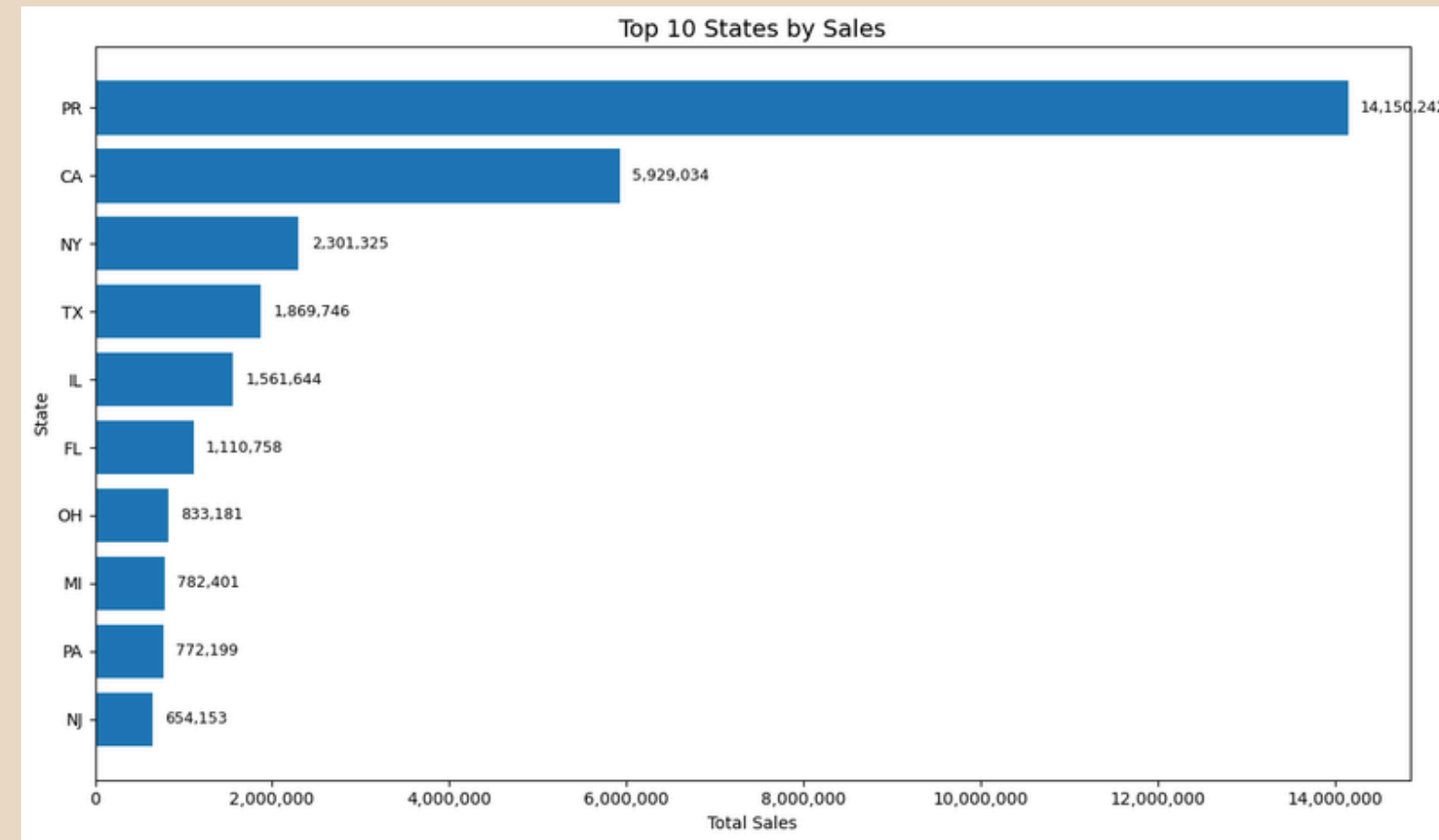
WHO BUYS FROM DATACO — AND WHERE?

Research Question 3

WHAT ARE THE PURCHASING PATTERNS OF DIFFERENT CUSTOMER SEGMENTS AND GEOGRAPHICAL REGIONS?



=> Consumer is the largest sales contributor



Puerto Rico → California → New York → Texas

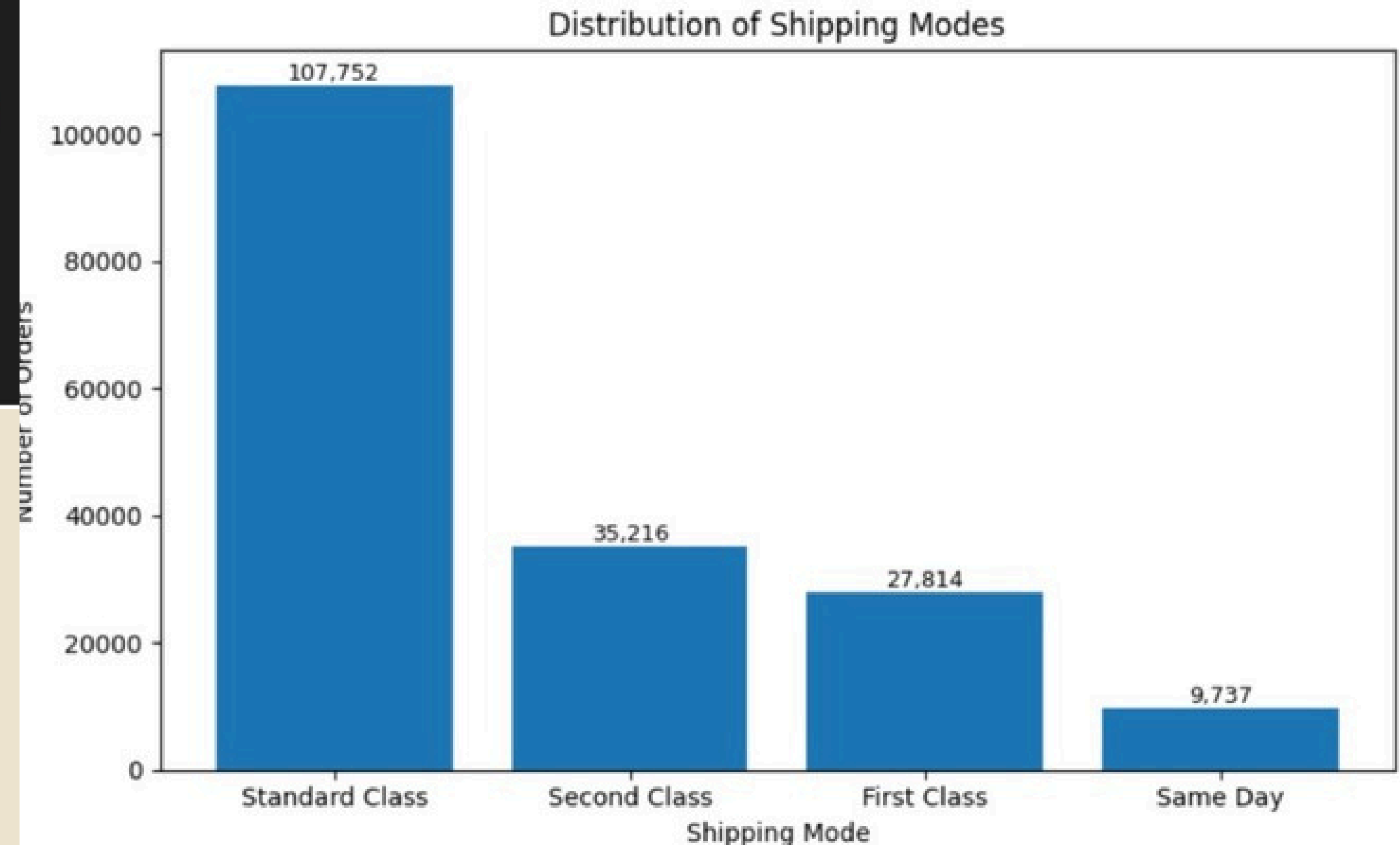
=> Sales are concentrated in a limited number of high-performing states.

KEY BUSINESS INSIGHT

Sales are concentrated in both a dominant customer segment and a limited number of high-performing markets.

Q4 How efficient is the company's logistics and delivery performance

```
SELECT
  o.shipping_mode,
  COUNT(oi.order_item_id) AS total_items
FROM OrderItems oi
JOIN Orders o
  ON oi.order_id = o.order_id
GROUP BY o.shipping_mode
ORDER BY total_items DESC;
```



Standard Class is the most frequently used shipping method

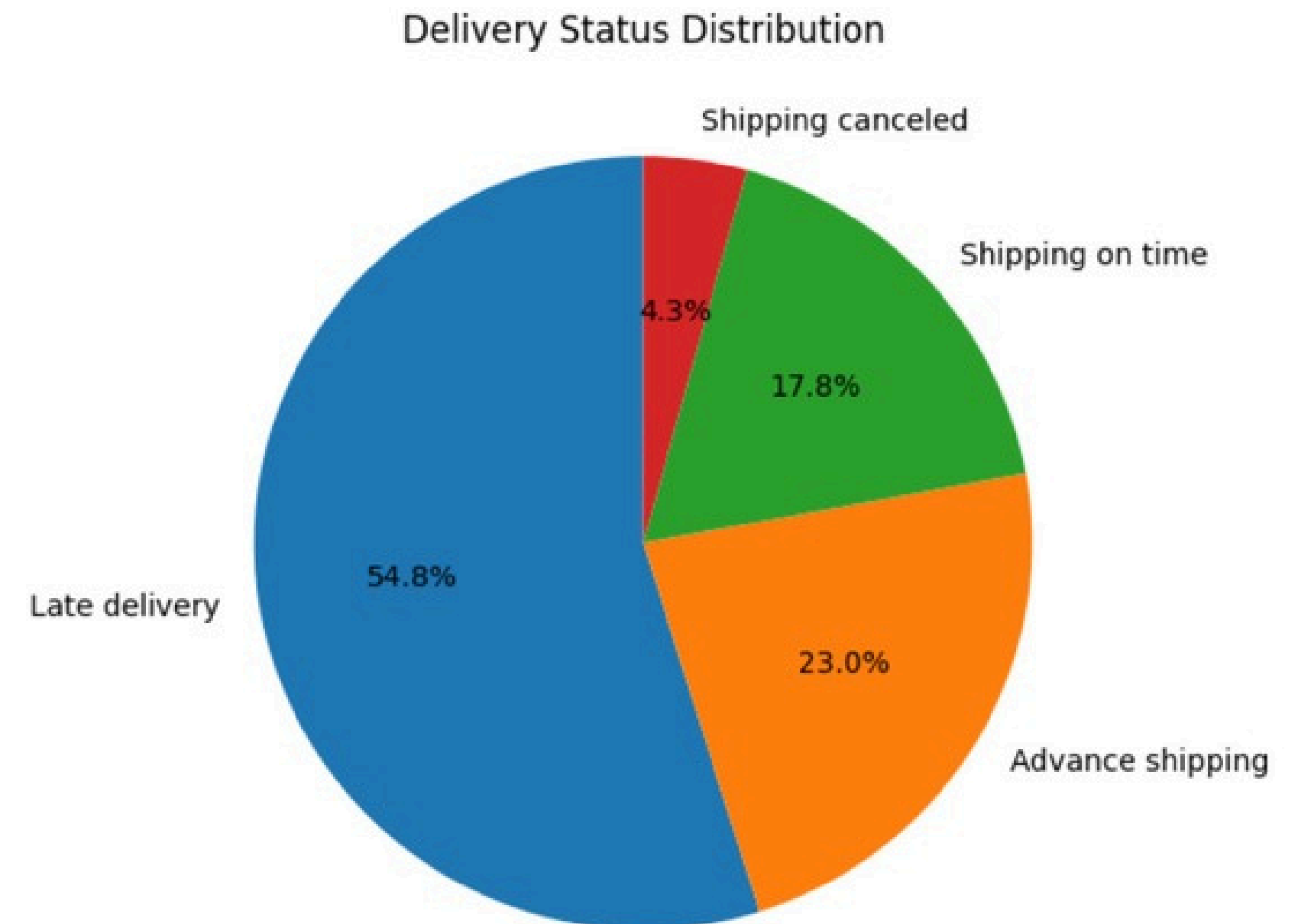
=> The overwhelming dominance of Standard Class indicates that price sensitivity outweighs delivery speed for the majority of retail orders (Consumer segment).
Customers generally prefer lower checkout costs over expedited shipping.

Q4 How efficient is the company's logistics and delivery performance

```
SELECT
    o.delivery_status,
    COUNT(oi.order_item_id) AS total_items
FROM OrderItems oi
JOIN Orders o
    ON oi.order_id = o.order_id
GROUP BY o.delivery_status
ORDER BY total_items DESC;
```

Delivery Status

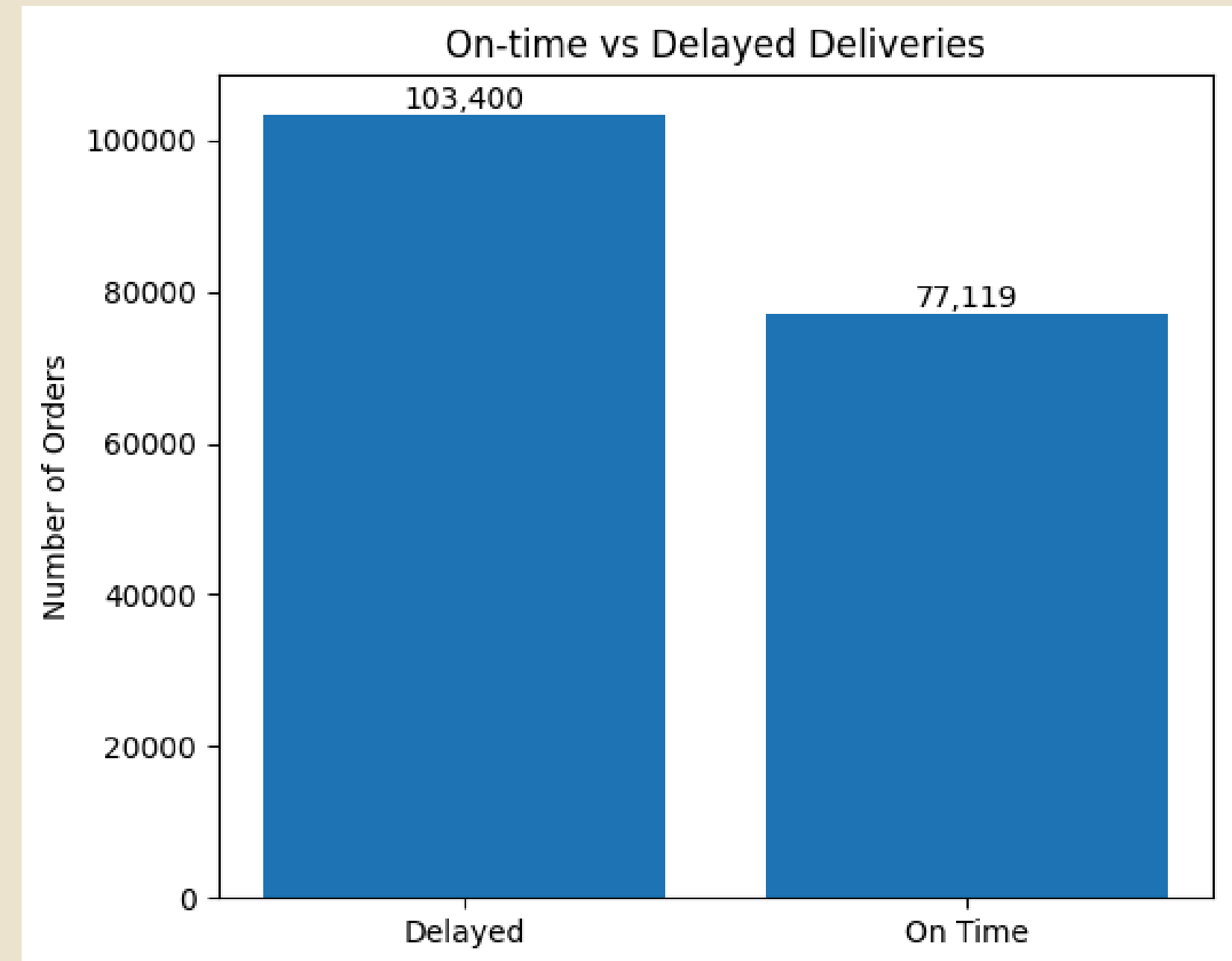
=> Over 55% of all orders are delivered late. This indicates a systemic operational failure across the fulfillment network rather than isolated carrier delays.



Q4 How efficient is the company's logistics and delivery performance

```
SELECT
  CASE
    WHEN oi.shipping_real_days <= oi.shipping_scheduled_days
      THEN 'On Time'
    ELSE 'Delayed'
  END AS delivery_performance,
  COUNT(*) AS total_orders
FROM OrderItems oi
GROUP BY delivery_performance
ORDER BY total_orders DESC;
```

On time vs Delayed Deliveries



Delayed deliveries occur more frequently than on-time deliveries

=> Negatively influence customer satisfaction, increase cancellation risk, and reduce repeat purchase behavior.

Q5

How do sales and profit change over time ?



- Stable performance: Monthly sales and profit remain relatively stable with moderate fluctuations, showing consistent customer demand and a stable revenue base.



- Sales–profit relationship: Profit generally follows sales trends, but lower–profit months may reflect higher costs, discounts, or product mix changes, highlighting the need for continuous monitoring and planning

4. CONCLUSION

Insight

- \$36.78M sales but only 10.8% margin
- Performance concentrated in a few categories
- Consumer several regions dominate sales
- Late deliveries exceed on-time deliveries
- Sales/profit are relatively stable

Recommendation

- Improve margins through pricing, discount, and product management.
- Prioritize high-performing products and review underperforming categories.
- Tailor customer and regional strategies to different market performance.
- Identify delivery bottlenecks and improve logistics reliability.
- Monitor costs and demand to optimize inventory and promotions.

THANK YOU